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CS1003
Programming in C

Module - 2

Content of Module-2

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- Data concepts in C:
  - Constants, Variables - Variables as named memory locations
  - Rules for Constructing Variables, understanding different Data Types - range, size, and appropriate usage,
  - C Keywords, format specifiers - %d, %c, %ud, %l, %f
  - Uses and Common Errors associated with them, Instruction, Pre-processor Directive, Macros.

- Operator in C:
  - Arithmetic Operators - Hierarchy of operations, Precedence
  - Relational Operators - (>, <, >=, <=, !=, ==)
  - Logical Operators - Logical AND (&&), Logical OR (||) Logical Not (!)
  - Conditional Operators - (?:),
  - Bit-wise Operators - (&, |, ^, ~, <<, >>),
  - Assignment Operators.
  - Type Conversion and Type Casting.

Constants

- Constant is a value that cannot be altered by the program during its execution.
- Used to represent fixed values that are intended to remain unchanged throughout the execution of the program

- Types of Constants
  - **Literal Constants:** Direct values in the code, such as numbers or characters.
    - Integer Literal: ‘10’,’-5’.
    - Floating Point Literal: ‘3.14’, ‘-0.678.
    - Character Literal: ‘a’,’z’.
    - String Literal:”RVUniversity”, “Hello World”

Variable

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- It is a storage location identified by a name (an identifier) that holds a value that can be modified during program execution.
- It must be declared before they can be used, specifying the type of data they will store.
- Types of Variable
  - Local Variable: Declared inside a function or block and can only be accessed within that function or block.
  - Global Variable: Declared outside of all functions and can be accessed from any function within the same file or other files

Variable

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```
#include <stdio.h>
int globalVar = 5; // Global variable
void exampleFunction() {
    int localVar = 10; // Local variable
    printf("Local Variable: %d\n", localVar);
}
int main() {
    int num = 20; // Local variable in main function
    printf("Global Variable: %d\n", globalVar);
printf("Local Variable in main: %d\n", num);
 exampleFunction();
 num = 30; // Modify local variable

 printf("Modified Local Variable in main: %d\n", num);
  return 0;
}
```

Rules for Constructing Variables

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- These rules ensure that variable names are valid and meaningful within the context of the C language
- Start with a Letter or Underscore:
  - Variable names must begin with a letter (either uppercase or lowercase) or an underscore(_)
  - Example: int count; float _value;
- Followed by Letters, Digits, or Underscores:
  - Example: int age1,double temperature_value;
- Cannot Start with a Digit: Example: int 1stPlace;
- Case-Sensitive: Example: int totalAmount; and int TotalAmount; are different variables
- Cannot Be a Reserved Keyword: Example: int int; is invalid because int is a keyword.
- No Special Characters Except Underscores: Example: int score$; is invalid.
- Length Limits: While C does not specify a strict limit on variable name length, many compilers have practical limits. However, it is advisable to keep names reasonably short and meaningful for better readability.

Understanding different Data Types - range, size, and appropriate usage

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- C data types are defined as the data storage format that a variable can store data to perform a specific operation.
- Datatypes are used to define a variable before to use in a program
- Uses
  - Identify the type of a variable when it declared.
  - Identify the type of the return value of a function
  - Identify the type of a parameter expected by a function

Why Data Types?

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- Every variable must have a type
- Every variable should get some space in the memory
- Help the compiler decide how many bits should be reserved for a variable

- Primary Data Types: These are fundamental or basic **data types in C** namely

1. int
2. float
3. character
4. void

Integer Data type

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- Integers are used to store whole numbers
- Unsigned -> means positive number if no sign i.e. +
- Signed -> means positive or negative number with sign i.e. +/-

Type	Size(bytes)	Range	Format Specifier
int or signed int	2	-32,768 to +32767	%d
unsigned int	2	0 to 65535	%d
long int or signed long int	4	-2,147,483,648 to +2,147,483,647	%ld
unsigned long int	4	0 to 4,294,967,295	%lu

Floating Point/Real Data type

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- Floating types are used to store real number

Type	Size(bytes)	Range	Format Specifiers
float	4	$3.4X10^{-38}$ to $3.4X10^{+38}$	%f
double	8	$1.7X10^{-308}$ to $1.7X10^{+308}$	%lf
long double	10	$3.4X10^{-4932}$ to $3.4X10^{+4932}$	%Lf

Character Data Type

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- Character types are used to store characters value

Type	Size(bytes)	Range	Format Specifiers
char or signed char	1	-128 to 127	%c
unsigned char	1	0 to 255	%c

Void Data Type

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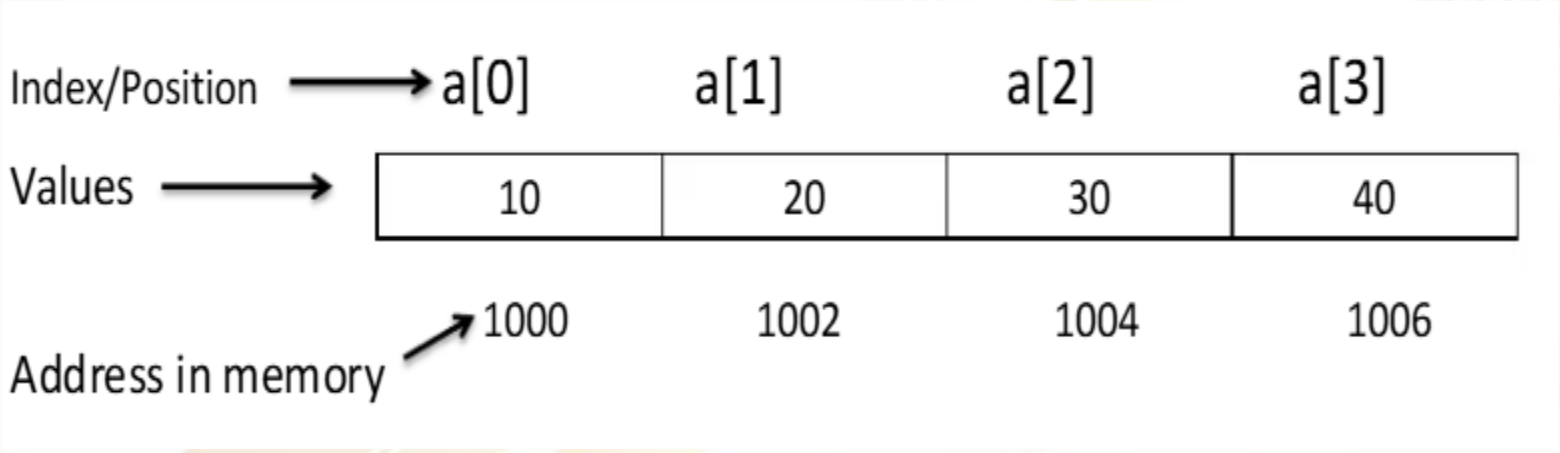
- Void type means no value.
- This is usually used to specify the type of functions which returns nothing by defining in front of main() or any other function and as blank arguments within functional brackets.

- For example
- Void main()
- Void main(void)

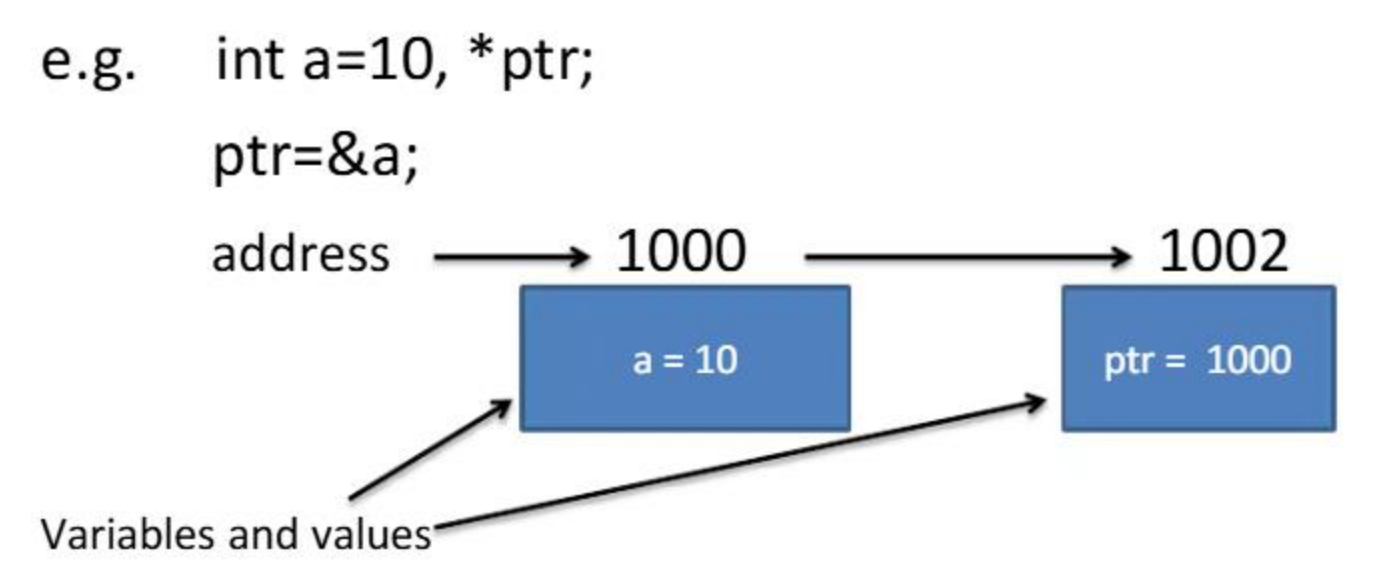
Derived Data Types

- Derived data types are types that are constructed from the basic (or primitive) data types.
- These derived types allow you to create complex data structures that can manage collections of data more effectively.
- Types
  - Arrays: Collection of items stored at continuous memory location
  - Example: int a[4]={10,20,30,40};



Derived Data Type cont..

- Pointers: Pointers are symbolic representation of addresses.



User-Defined Data Types

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- User-defined data types are data types that you define yourself, as opposed to the built-in primitive data types like int, char, and float.
- User-defined data types allow you to create complex data structures that better represent real-world entities and manage data in a more organized manner
- structure
- union
- enum
- typedef

Data Type	Size	Range (Typical)	Usage
`char`	1 byte (8 bits)	Signed: -128 to 127 Unsigned: 0 to 255	Store single characters or small integers.
`short`	2 bytes (16 bits)	Signed: -32,768 to 32,767 Unsigned: 0 to 65,535	Store smaller integers.
`int`	4 bytes (32 bits)	Signed: -2,147,483,648 to 2,147,483,647 Unsigned: 0 to 4,294,967,295	General-purpose integer storage.
`long`	4 bytes (32 bits) or 8 bytes (64 bits)	Signed (32-bit): -2,147,483,648 to 2,147,483,647 Unsigned (32-bit): 0 to 4,294,967,295 Signed (64-bit): -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 Unsigned (64-bit): 0 to 18,446,744,073,709,551,615	Larger integer storage when `int` is insufficient.
`long long`	8 bytes (64 bits)	Signed: -9,223,372,036,854,775,808 to 9,223,372,036,854,775,807 Unsigned: 0 to 18,446,744,073,709,551,615	Very large integer storage.
`float`	4 bytes (32 bits)	±1.7E−38 to ±3.4E+38	Single precision floating-point numbers.
`double`	8 bytes (64 bits)	±1.7E−308 to ±1.7E+308	Double precision floating-point numbers.
`long double`	8 bytes (64 bits), 12 bytes (96 bits), or 16 bytes (128 bits)	Typically larger than `double`, varies by implementation	Extended precision floating-point numbers.
`_Bool` (C99)	1 byte (8 bits)	`true` or `false`	Boolean values.

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Derived Data Type	Definition	Example	Usage
Array	Collection of elements of the same type stored in contiguous memory locations.	`int arr[10];`	To store multiple values of the same type.
Structure (`struct`)	User-defined type that groups variables of different types under a single name.	`struct Person { char name[50]; int age; };`	To group related variables together.
Union (`union`)	User-defined type where all members share the same memory location.	`union Data { int i; float f; };`	To store different types of data in the same memory location.
Enumeration (`enum`)	User-defined type consisting of integral constants.	`enum Weekday { MONDAY, TUESDAY, WEDNESDAY };`	To define a set of named integer constants.
Pointer	Variable that stores the address of another variable.	`int *ptr;`	To reference and manipulate memory addresses.

Keywords in C

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- Keywords are reserved words that have special meanings and cannot be used for identifiers (e.g., variable names, function names).
- They are part of the C language syntax and are predefined by the language specification.
- Each keyword serves a specific purpose in C, defining the language's structure and control flow.

Keyword	Description
auto	Specifies automatic storage duration (default storage class for local variables).
break	Exits from a loop or a switch statement.
case	Defines a branch in a switch statement.
char	Declares a variable of character type.
const	Specifies that the value of a variable cannot be changed
continue	Skips the current iteration of a loop and proceeds to the next iteration.
default	Specifies the default case in a switch statement.
do	Initiates a do-while loop, which executes a block of code at least once and then repeats based on a condition.
double	Declares a variable of double precision floating-point type.
else	Specifies the block of code to execute if the condition in an if statement is false.
enum	Defines an enumeration, a set of named integer constants.
extern	Declares a variable or function that is defined in another file or scope.
float	Declares a variable of floating-point type.
for	Initiates a for loop, which repeats a block of code a specified number of times.

Keyword	Description
goto	Transfers control to a labeled statement within the same function.
if	Starts a conditional statement that executes a block of code if a specified condition is true.
int	Declares a variable of integer type.
long	Declares a variable of long integer type.
register	Suggests that the variable should be stored in a CPU register for faster access (note: this is a suggestion, not a guarantee).
return	Exits from a function and optionally returns a value.
short	Declares a variable of short integer type.
signed	Specifies a signed integer type, which can hold both positive and negative values.
sizeof	Returns the size, in bytes, of a variable or data type.
static	Specifies that the variable or function has internal linkage or maintains its value between function calls.
struct	Defines a structure, a composite data type that groups variables of different types.
switch	Allows multi-way branching based on the value of an expression.
typedef	Creates an alias for a data type.
union	Defines a union, a data type that can hold different types of data but only one at a time.

Keyword	Description
unsigned	Specifies an unsigned integer type, which can only hold non-negative values.
void	Specifies that a function does not return a value or indicates a generic pointer type.
volatile	Indicates that a variable may be changed by external factors (e.g., hardware) and should not be optimized by the compiler.
while	Initiates a while loop, which repeats a block of code as long as a specified condition is true.

C format specifiers - %d, %c, %ud, %l, %f, Uses and

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- format specifiers are used in functions like printf and scanf to define how data should be formatted when output or input is performed
- Format Specifiers(%d)
  - Use: Formats an int (signed integer) for output.
  - Example:
```
int num = 42;
printf("%d", num); // Output: 42
```
  - Using %d with a type other than int can lead to undefined behavior.
  - For instance, using %d with unsigned int can result in incorrect output.

- Format Specifier(%c)
- **Use**: Formats a single character for output.
- Example: char ch = 'A';

  printf("%c", ch); // Output: A
- Using %c with a non-character type, such as an int, without explicit casting, can produce unexpected results.

- Format specifier(%u)
- Use: Formats an unsigned int for output.
- Example: unsigned int num = 42;

  printf("%u", num); // Output: 42
- Using %u with a signed integer or other non-unsigned types can cause incorrect output or undefined behavior

- Format Specifier(%ld)
- **Use**: Formats a long int (signed long integer) for output.
- Example: long int num = 123456789L;

  printf("%ld", num); // Output: 123456789
- Using %ld with int or short can cause incorrect output. Ensure the type matches the format specifier.
- Format specifier(%f)
- Use: Formats a float or double for output. When using %f, it assumes a double argument due to default argument promotion
- Example: float num = 3.14f;

  printf("%f", num); // Output: 3.140000
- Using %f with types other than float or double can result in incorrect output or undefined behavior. Also, note that %f will print a float with six decimal places by default.

Format Specifier Common Errors associated with them

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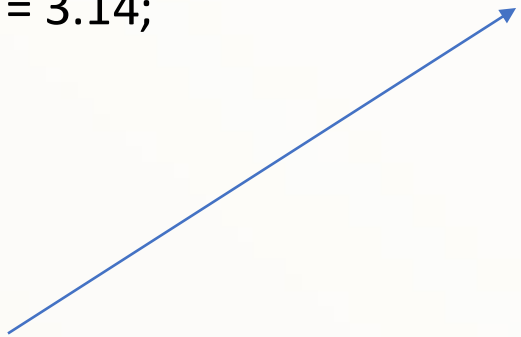
- **Type Mismatch**:
  - Using a format specifier that does not match the type of the argument leads to incorrect output or undefined behavior.
  - For example, using %d for a float or %f for an int is incorrect.
- Precision and width Misuse
  - For floating-point numbers, if you need to control the number of decimal places, use %.nf where n is the number of decimal places.
  - For example, %.2f formats a float to two decimal places.
- Unsigned Vs Signed Confusion
  - Using %u for signed integers or %d for unsigned integers can lead to confusing and incorrect output.
  - Ensure you match the specifier to the type (e.g., %u for unsigned int, %d for int).
- Long and short Modifiers
  - When using %ld, %lld, %hd, etc., ensure that the variable type matches the specifier.
  - %ld should be used for long int, %lld for long long int, and %hd for short int.

- Floating point precision
  - The %f specifier shows six decimal places by default.
  - If more precision is needed or if you want to round off to a specific number of decimal places, adjust accordingly.
- Proper use of format specifiers is essential for accurate and predictable behavior in C programs.
- Always match your format specifiers with the type of the variables being used, and double-check to avoid common pitfalls

Instructions in C

- **Instructions** in C are the fundamental building blocks of a C program.
  - They include statements that perform actions such as

1. Variable declarations: Ex: int a; float b;
2. Assignments: Ex: a = 10; b = 3.14;
3. Control flow operations. :Ex



Pre-Processor Directives in C

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- Pre-processor directives are instructions processed by the pre-processor before actual compilation begins.
- Start with a # symbol and are used to control the compilation process.
- Some common pre-processor directives include:
- #include: Includes the contents of a file in the source code.
  - #include <stdio.h> // Includes the standard input/output library
- #define: Defines macros or constants. This directive replaces all instances of the macro with its definition before compilation.
  - #define PI 3.14159 // Defines a constant PI with the value 3.14159
- #undef: Undefines a macro, removing its definition.
  - #undef PI // Undefines the macro PI

Pre-Processor Directives in C

- #ifdef and #ifndef: Checks if a macro is defined or not defined, respectively. Useful for conditional compilation.
  - #ifdef DEBUG printf("Debug mode is enabled\n"); #endif
- #if, #elif, and #else: Conditional compilation based on compile-time constants.

```
#if defined(DEBUG) && (DEBUG > 1)
        printf("Debug level 2 or higher\n");
#else
        printf("Debug level 1 or not defined\n"); #endif
```

- #endif: Ends an #if, #ifdef, or #ifndef block.
- #error: Generates a compilation error with a specified message.
  - #error "This is an error message”
- #pragma: Provides additional information to the compiler; usage is compiler-specific.
  - #pragma once // Ensures the header file is included only once

Macros in C

- Macros are a type of pre-processor directive that allows you to define code snippets or constants that can be reused throughout your program.
- Macros are defined using #define and can be simple constants or more complex code blocks.

- Object-like Macros: Define constants or values.

```
#define MAX_SIZE 100 // Defines MAX_SIZE as 100
```

- **Function-like Macros**: Define code snippets that can take arguments.

```
#define SQUARE(x) ((x) * (x)) // Defines a macro to compute the square of a number
```

Ex: int result = SQUARE(5); // Expands to ((5) * (5)) and evaluates to 25

Pre defined Macros

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Macro	Value
__DATE__	A string containing the current date.
__FILE__	A string containing the file name.
__LINE__	An integer representing the current line number.
__STDC__	If follows ANSI standard C, then the value is a nonzero integer.
__TIME__	A string containing the current time.

Pre defined Macros_ Examples

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```
#include<stdio.h>
int main()
{
printf("Date is %s\n",__DATE__);
printf("Time is %s\n",__TIME__);
printf("File is %s\n",__FILE__);
printf("Line is %d\n",__LINE__);
printf("ANSI is %d\n",__STDC__);
return 0;
}
```

Operators in C: Arithmetic operators

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- Used to perform mathematical operations.
- Understanding their hierarchy, associativity, and precedence is crucial for writing correct expressions and avoiding errors.
- Operator precedence determines the order in which operators are evaluated in an expression

1. Parentheses ():Operations inside parentheses are performed first.Example: (2 + 3) * 4 evaluates to 5 * 4, which is 20.
2. Unary Operators:
    1. Unary plus + and unary minus - (e.g., -5, +3)
    2. Increment ++ (e.g., x++, ++x)
    3. Decrement -- (e.g., x--, --x)
    4. These are evaluated after parentheses but before other arithmetic operations

Arithmetic operators cont..

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3. Multiplication, Division, and Modulus *, /, %:
   - Same level of precedence and are evaluated from left to right.
   - Example: 10 / 2 * 3 evaluates to 5 * 3, which is 15.

4. Addition and Subtraction +, -
   - Same level of precedence and are evaluated from left to right.
   - Example: 5 + 3 - 2 evaluates to 8 - 2, which is 6.

Associativity

- Associativity determines the order in which operators of the same precedence are evaluated:

1. Left-to-Right Associativity:
   1. Operators like *, /, %, +, and - are evaluated from left to right.
   2. Example: 6 / 2 * 3 evaluates as (6 / 2) * 3, which is 3 * 3, giving 9.

2. Right-to-Left Associativity:

- Unary operators like ++, --, and unary + and - have right-to-left associativity.
- Example: --x + 5 first decrements x and then adds 5.

Operator	Description	Precedence	Associativity
()	Parentheses	Highest	N/A
++ (post)	Post-increment	High	Left-to-Right
-- (post)	Post-decrement	High	Left-to-Right
++ (pre)	Pre-increment	High	Right-to-Left
-- (pre)	Pre-decrement	High	Right-to-Left
+ (unary)	Unary plus	High	Right-to-Left
- (unary)	Unary minus	High	Right-to-Left
*	Multiplication	Medium	Left-to-Right
/	Division	Medium	Left-to-Right
%	Modulus	Medium	Left-to-Right
+	Addition	Low	Left-to-Right
-	Subtraction	Low	Left-to-Right

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- Examples

- 3 + 5 * 2 evaluates as
  - 3 + (5 * 2), which is 3 + 10,
  - Resulting in 13.

- (3 + 5) * 2
  - Evaluates as 8 * 2,
  - which is 16.

- Examples

① `8 / 4 * 16`

`2 * 16`

`32`

② `8 * 4 / 16`

`32 / 16`

`2`

③ Assume a = 8, b = 15, c = 4.

`a * ((a%5) * (4 + (b-3)/(c+2)))`

`2 * ((8%5) * (4 + (15-3)/(4+2)))`

- Parentheses will have the highest priority and then next priority is for the operators.

`2 * ( 3 * (4 + (15-3)/(4+2)))`

`2 * (3 * (4 + 12 / (4+2)))`

`2 * (3 * (4 + 12/6))`

`2 * (3 * (4 + 2))`

`2 * (3 * 6)`

`2 * 18`

`36`

- Example

(4) Assume a=1, b=-5, c=6. Evaluate the following expression x1 = (-b+sqrt(b*b-4*a*c))/(2*a)

Soln: After Substituting

x1 = (-(-5) + sqrt((-5)*(-5)-4*1*6))/(2*1)

x1 = (5 + sqrt((-5)*(-5)-4*1*6))/(2*1)

x1 = (5 + sqrt(25 - 4*1*6))/(2*1)

x1 = (5 + sqrt(25 - 4*6))/(2*1)

x1 = (5 + sqrt(25 - 24))/(2*1)

x1 = (5 + sqrt(1))/(2*1)

x1 = (5 + 1.0)/(2*1)

x1 = 6.0/(2*1)

x1 = 6.0/2

x1 = 3.0

Relational Operators

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- Relational operators are used to compare values in expressions

Description	Operator	Priority	Associativity
Less than	<	1	Left to Right
Less than/equal	<=	1	Left to Right
Greater than	>	1	Left to Right
Greater than/equal	>=	1	Left to Right
Equal	==	2	Left to Right
Not Equal	!=	2	Left to Right

Equality Operators

Rules to follow while evaluating Relational Expression

- Evaluate Expression within parenthesis
- Evaluate Unary operators
- Evaluate Arithmetic Expression
- Evaluate Relational Expression

Evaluate the following expression

100 /20< = 10 - 5 + 100 % 10 - 20 == 5 >= 1 ! = 20

- No expression inside parenthesis and no unary operators.
- Arithmetic operators are evaluated first next relational

5 < = 10 - 5 + 100 % 10 - 20 == 5 >= 1 ! = 20

5 < = 10 - 5 + 0 - 20 == 5 >= 1 ! = 20

5 < = 5 + 0 - 20 == 5 >= 1 ! = 20

5 < = 5 - 20 == 5 >= 1 ! = 20

5 < = -15 == 5 >= 1 ! = 20

0 == 5 >= 1 ! = 20

0 == 1 ! = 20

0 ! = 20

1

Result = 1

Logical Operators

- Logical operators are used to perform logical operations on boolean values.
- Essential for controlling the flow of a program, especially in conditional statements and loop
- Used to combine 2 or more relational expressions are called logical operations
- Expression contains only logical operators are called logical expression

Description	Operator	Priority	Associativity
NOT(Unary Operator)	!	1	Left to Right
AND(Binary)	&&	2	Left to Right
OR(Binary)	||	3	Left to Right

Logical NOT(!)

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Operand	!Operand
True(1)	False(0)
False(0)	True(1)

Logical AND(&&)

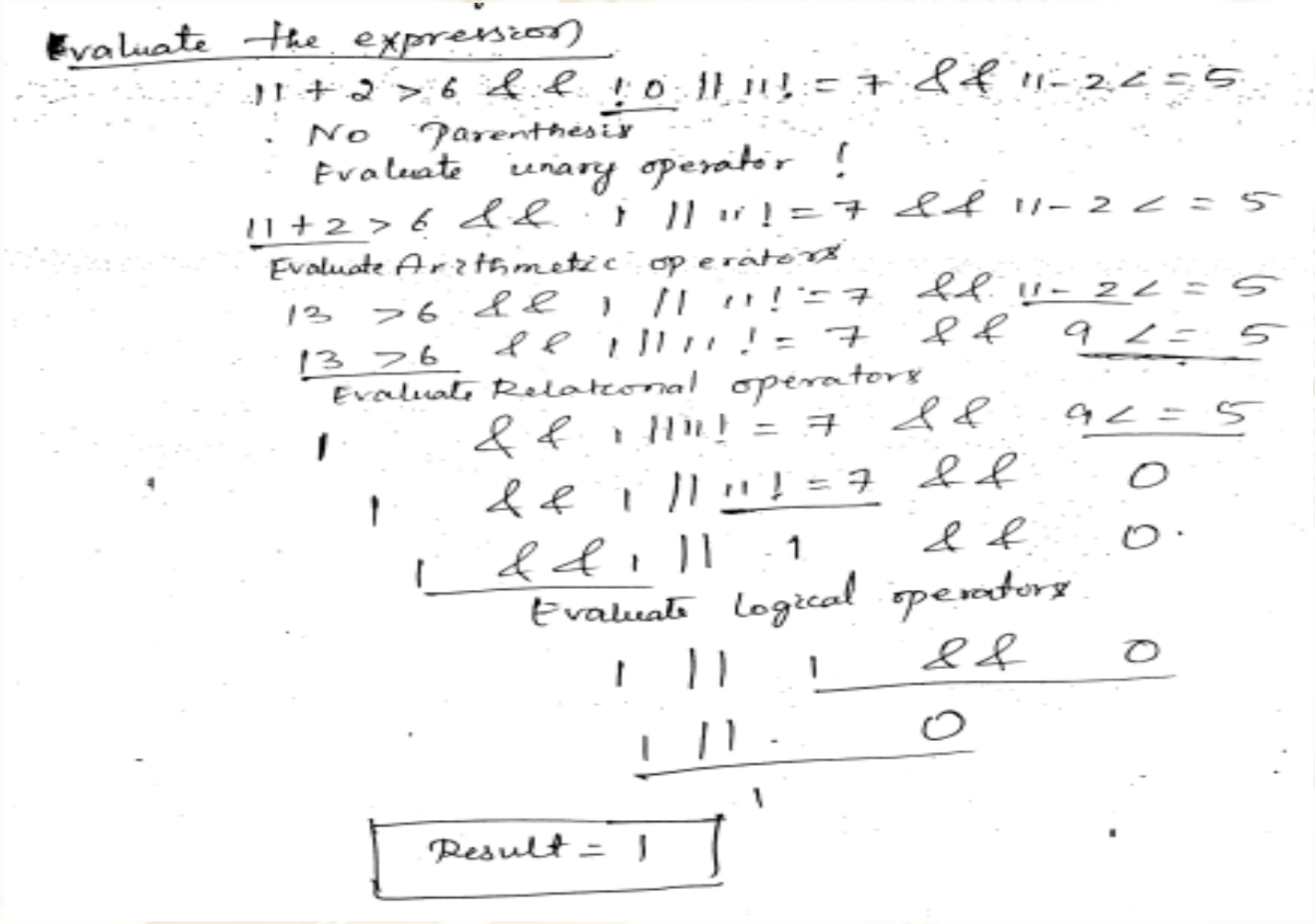
Operand1	AND	Operand2	Result
True(1)	&&	True(1)	True(1)
True(1)	&&	False(0)	False(0)
False(0)	&&	True(1)	False(0)
False(0)	&&	False(0)	False(0)

Logical OR(II)

Operand1	OR	Operand2	Result
True(1)	||	True(1)	True(1)
True(1)	||	False(0)	True(1)
False(0)	||	True(1)	True(1)
False(0)	||	False(0)	False(0)

Rules to be follow to evaluate Logical Expression

- Evaluate Expression within parenthesis
- Evaluate Unary operators
- Evaluate Arithmetic Expression
- Evaluate Relational Expression
- Evaluate Logical Expression



Conditional Operators.

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- Make decisions based on conditions.
- Used within expressions to choose between two values based on a condition.
- The most common conditional operator is the ternary operator
- Ternary conditional operator is a compact way to express a simple if-else statement in a single line.
- It takes three operands and has the following syntax
  - condition ? expression_if_true : expression_if_false;
  - condition: An expression that evaluates to true or false.
  - expression_if_true: The expression that is evaluated and returned if the condition is true.
  - expression_if_false: The expression that is evaluated and returned if the condition is false.

Example

```
#include <stdio.h>
int main()
{
int a = 10, b = 20;
int max = (a > b) ? a : b; // max will be 20
printf("Max value is %d\n", max);
return 0;
}
```

Operator Precedence and Associativity

- **Precedence**:
  - The ternary operator (? :) has higher precedence than assignment operators but lower than most arithmetic operators.
- **Associativity**: Right-to-Left

- Program to find largest of 3 nos

```
#include <stdio.h>
int main()
{
int a, b, c, largest;
printf("Enter three numbers: ");
scanf("%d %d %d", &a, &b, &c);
largest=(a>b)?((a>c)?a:c)(b>c)?b:c);
printf("The largest number is= %d\n", largest);

    return 0;
}
```

Bitwise Operators

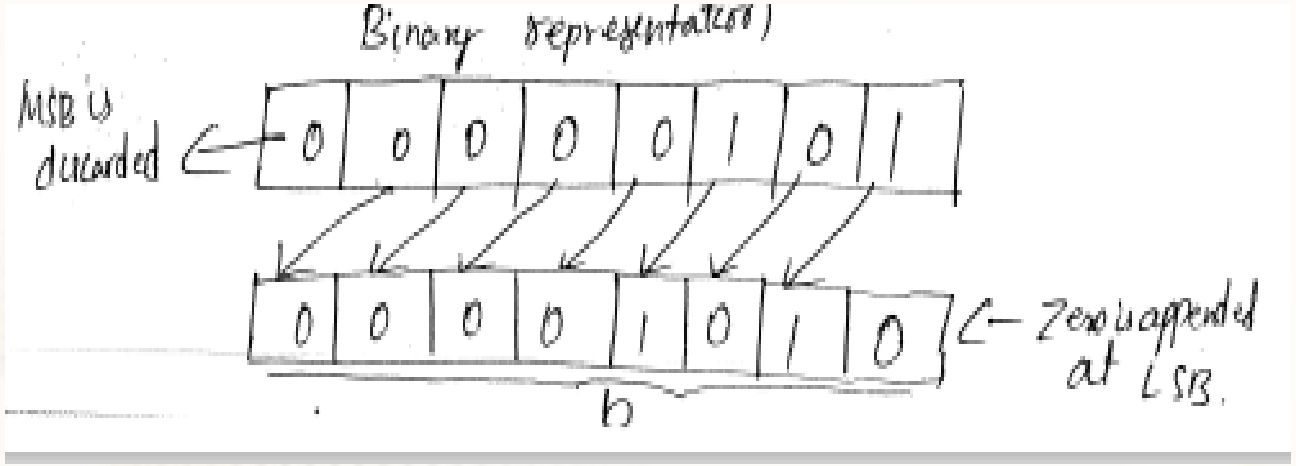
- Bitwise operators perform operations on binary representations of integers

Descriptions	Operator	Priority	Associativity
Bit-wise Negate	~	1	Left to right
Left Shift	<<	2	Left to right
Right Shift	>>	2	Left to right
Bit-wise and	&	3	Left to right
Bit-wise xor	^	4	Left to right
Bit-wise or	|	5	Left to right

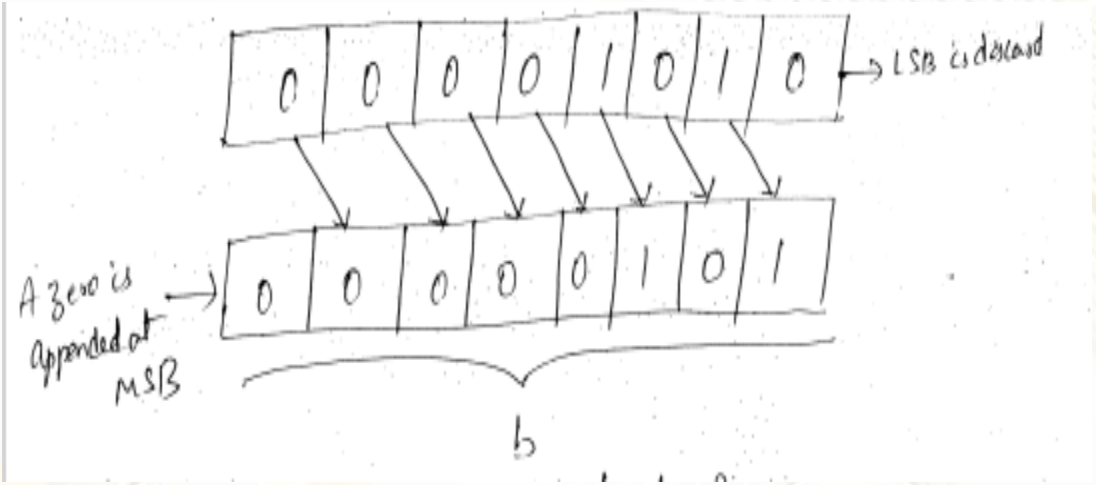
- Bit-wise Negate(~) (one's complement)
  - Inverts all the bits of the operand.
  - This means each 1 becomes 0, and each 0 becomes 1.
  - Example: ~5 --□ 0101 (5 in binary) , result□ 1010

Left and Right shift Operators

- Shifts the bits of the first operand to the left by the number of positions specified by the second operand and vice versa for right shift.
- The vacated positions on the right are filled with 0.
- Example: b=5 << 1



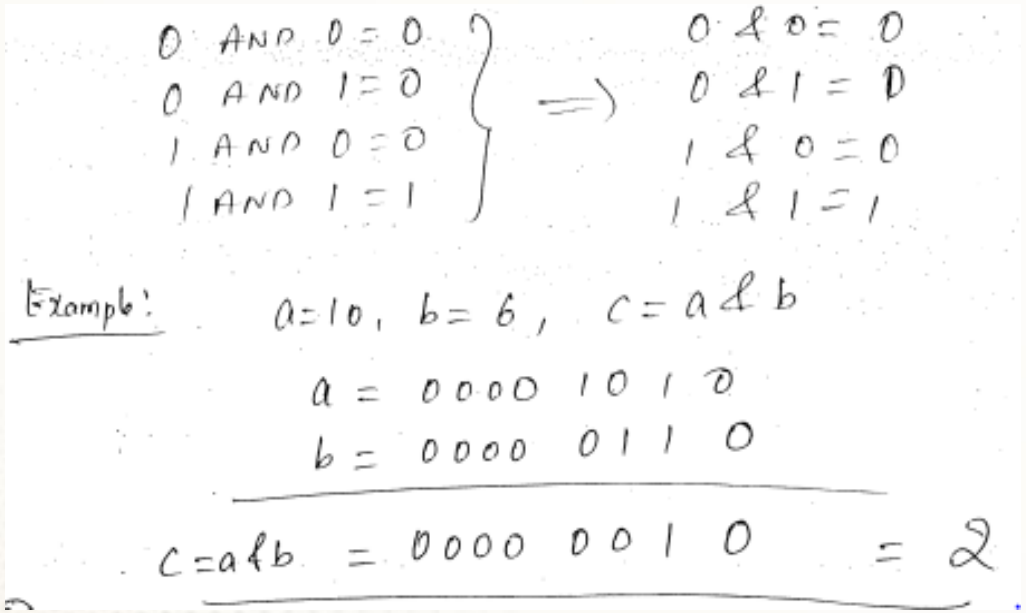
b=10>>1

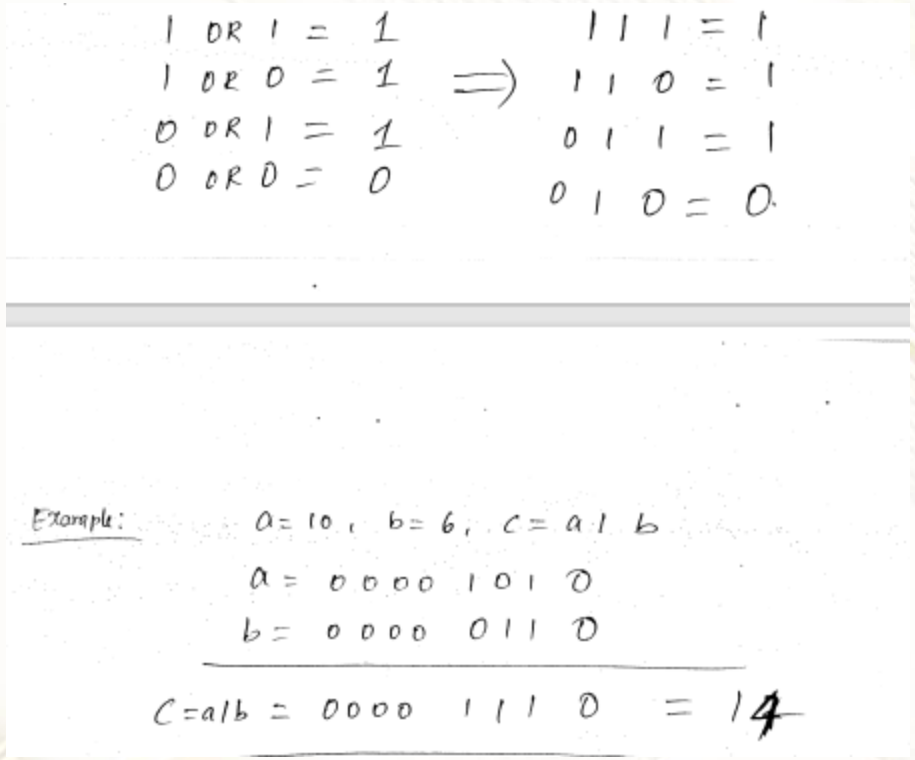


- Shift towards left by 1 bit is equivalent to multiplying by 2
- Shift towards Right by 1 bit is equivalent to divide by 2

Bit-wise And & Bit-wise Or

- Each bit in the result is 1 if the corresponding bits of both operands are 1,otherwise it is 0.





- Each bit in the result is 1 if at least one of the corresponding bits of the operands is 1

Bit-wise XOR

- It is represented by the caret symbol (^).
- It performs a bitwise exclusive OR operation between two integer values.

Example

```
#include <stdio.h>
int main() {
unsigned int a = 12;  // Binary: 00001100
unsigned int b = 7;   // Binary: 00000111
unsigned int result = a ^ b; // Binary result: 00001011
printf("a = %u (Binary: %b)\n", a, a);
printf("b = %u (Binary: %b)\n", b, b);
printf("a ^ b = %u (Binary: %b)\n", result, result);return 0;
}
```

A	B	A^B
0	0	0
0	1	1
1	0	1
1	1	0

Assignment Operators

- Assignment operators are used to assign values to variables.
- Most commonly used assignment operator is the basic =
- There are several compound assignment operators that combine an operation with an assignment
- Basic Assignment(=) Operator
  - Assigns the value on the right to the variable on the left.
  - Example: Int a=10;
- Compound Assignment Operators
  - These operators combine an arithmetic or bitwise operation with assignment

+=	-=	*=	/=	%=	&=	|=	^=
int a=10	int a=10	int a=10	int a=10	int a=10	int a=12, binary:00001100	int a=12, binary:00001100	int a=12, binary:00001100
a+=3 a=a+3	a-=3 a=a-3	a*=3 a=a*3	a/=3 a=a/3	a%=3 a=a%3	int b=7,binary:00000111	int b=7,binary:00000111	int b=7,binary:00000111
a=13	a=7	a=30	a=3	a=1	a &= b; Result: 00000100 (Binary), a now holds 4	a |= b; // Result: 00001111 (Binary), a now holds 15	a ^= b; // Result: 00001011 (Binary), a now

Cont..

- Basic Assignment: =
- Arithmetic Assignment: +=, -=, *=, /=, %=
- Bitwise Assignment: &=, |=, ^=, <<=, >>=

<<=	>>=
int a=5;	int a = 20;
Binary: 00000101	// Binary: 00010100
a <<= 1; // Result: 00001010 (Binary), a now holds 10	a >>= 2; // Result: 00000101 (Binary), a now holds 5

Type Casting and Type Conversion

- **Type conversion** and **type casting** are two related but distinct concepts used to handle and manipulate data type.
- **Type conversion** refers to the automatic or implicit conversion of data from one type to another by the compiler.
- This happens when the compiler converts data types as part of an expression evaluation to ensure the operation is performed correctly.
- Implicit Type Conversion
  - When performing arithmetic operations involving mixed types, the compiler automatically converts the operands to a common type, usually the type with the highest precision.
  - When assigning a value of a smaller data type to a larger data type, the conversion happens implicitly.

Ex: `int a = 5;  float b = 4.2;`

`float result = a + b;` // 'a' is implicitly converted to float // 'result' will be 9.2 (float)

Type Casting

- **Type casting** is a method used to explicitly convert a variable from one type to another.
- This is done using a cast operator and allows you to control the conversion process more precisely.
- **Syntax**: (target_type) expression
- Example:**Casting Between Numeric Types:** When you need to explicitly convert one numeric type to another, you use type casting.

```
int a = 5; float b = 2.3;

float result = (float) a / b; // Cast 'a' to float before division // 'result' will be 2.173913 (float)
```

Cont...

- **Casting to a Smaller Type:**
  - Casting from a larger to a smaller type may lead to loss of data.

Example

```
double a = 9.99;
int b = (int) a;  // Cast 'a' to int // 'b' will be 9, fractional part .99 is discarded
```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